Physics 468/668: Statistical Physics **Spring 2019**

Lecture 16: Van der Waals equation from the Perturbation theory

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16.1 Perturbation Theory

Recall that the configuration partition function:

$$Z_N = \int \exp(-\beta U) dr_1 \cdots dr_N \tag{16.1}$$

Now we divide U to two source of contributions

$$U = U_0 + U_1, \tag{16.2}$$

where U_1 is small compared to U_0.

We first compute Z due to U_0,

$$Z_N^0 = \int \exp(-\beta U_0) dr_1 \cdots dr_N \tag{16.3}$$

Then, we express Z_N as

$$Z_N = \int \exp(-\beta U) dr_1 \cdots dr_N = Z_N^0 \langle \exp(-\beta U_1) \rangle_0 \tag{16.4}$$

The total partition function is:

$$Q(N, V, T) = \frac{Z(N, V, T)}{N! \lambda^{3N}} \tag{16.5}$$

The free energy is given by

$$A(N, V, T) = -\frac{1}{\beta} \ln \left(\frac{Z_N}{N! \lambda^{3N}} \right) = -\frac{1}{\beta} \left(\frac{Z_N^0}{N! \lambda^{3N}} \right) - \frac{1}{\beta} \ln \langle \exp(-\beta U_1) \rangle_0 \tag{16.6}$$

We hence separate A into two terms

$$\begin{aligned} A^0(N, V, T) &= -\frac{1}{\beta} \left(\frac{Z_N^0}{N! \lambda^{3N}} \right) \\ A^1(N, V, T) &= -\frac{1}{\beta} \ln \langle \exp(-\beta U_1) \rangle_0 = -\frac{1}{\beta} \ln \sum_{k=0}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \end{aligned} \tag{16.7}$$

Where, the 2nd term in the 2nd expression includes a power series expansion of exponential function.

Here we hope to expand A^1 in a general polynomial form,

$$A^1 = \sum_{k=1}^{\infty} \frac{(-\beta)^{k-1}}{k!} w_k \tag{16.8}$$

where w_k are a set of coefficients to be determined.

Using the fact that

$$\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{x^k}{k} \tag{16.9}$$

We have that

$$\begin{aligned} -\frac{1}{\beta} \ln\left(\sum_{k=0}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \right) &= -\frac{1}{\beta} \ln\left(1 + \sum_{k=1}^{\infty} \frac{(-\beta)^k}{k!} \langle U_1^k \rangle_0 \right) \\ &= -\frac{1}{\beta} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} \left(\sum_{l=1}^{\infty} \frac{(-\beta)^l}{l!} \langle U_1^l \rangle_0 \right)^k \end{aligned} \tag{16.10}$$

Comparing with Eq. 16.8

$$-\frac{1}{\beta} \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k} \left(\sum_{l=1}^{\infty} \frac{(-\beta)^l}{l!} \langle U_1^l \rangle_0 \right)^k = \sum_{k=1}^{\infty} \frac{(-\beta)^{k-1}}{k!} w_k \tag{16.11}$$

From the β^1 term, we find

$$-\frac{\beta \langle U_1 \rangle_0}{1!} = -\frac{\beta w_1}{1!} \tag{16.12}$$

so

$$w_1 = \langle U_1 \rangle_0 \tag{16.13}$$

Likewise, we get

$$w_2 = \langle U_1^2 \rangle_0 \langle U_1 \rangle_0^2 \tag{16.14}$$

$$w_3 = \langle U_1^3 \rangle_0 + 2\langle U_1 \rangle_0^3 - 3\langle U_1 \rangle_0 \langle U_1^2 \rangle_0 \tag{16.15}$$

Therefore, the free energy can be expressed as

$$\begin{aligned} A &= A^0 + w_1 - \frac{\beta}{2} w_2 + \frac{\beta^2}{6} w_3 \cdots \\ &= -\frac{1}{\beta} \ln\left(\frac{Z_N^0}{N! \lambda^{3N}} \right) + \langle U_1 \rangle_0 - \frac{\beta}{2} (\langle U_1^2 \rangle_0 \langle U_1 \rangle_0^2) + \frac{\beta^2}{6} (\langle U_1^3 \rangle_0 + 2\langle U_1 \rangle_0^3 - 3\langle U_1 \rangle_0 \langle U_1^2 \rangle_0) \cdots \end{aligned} \tag{16.16}$$

Now the next task is to obtain some analytical formula for $\langle U_1 \rangle_0$, suppose that U_1 adopts a pair potential

$$U_1 = \frac{1}{2} \sum_{i \neq j} u_1(r_{ij}) \tag{16.17}$$

Then

$$\begin{aligned}\langle U_1\rangle_0 &= \frac{1}{Z_N^0}\int \frac{1}{2}\sum_{i\neq j} u_1(r_{ij})\exp(-\beta U_0)dr_1\cdots dr_N \\ &= \frac{N(N-1)}{2Z_N^0}\iint dr_1 dr_2 u_1(r_{12})\int\cdots\int \exp(-\beta U_0)dr_3\cdots dr_N \\ &\approx \frac{N^2}{2V^2}\iint dr_1 dr_2 u(r_{12})g(r_1,r_2) \\ &= \frac{\rho^2 V}{2}\int_0^\infty 4\pi r^2 u_1(r)g(r)dr\end{aligned} \tag{16.18}$$

where $\rho = N/V$,
$g(r)$ is the pair distribution function https://en.wikipedia.org/wiki/Radial_distribution_function:

$$g(r) = g(r_{12}) = g(r_1,r_2) = \frac{V^2(N-1)}{NZ_N}\int\cdots\int \exp(-\beta U)dr_3\cdots dr_N \tag{16.19}$$

So the free energy is

$$A = -\frac{1}{\beta}\ln\left(\frac{Z_N^0}{N!\lambda^{3N}}\right) + \frac{\rho^2 V}{2}\int_0^\infty 4\pi r^2 u_1(r)g(r)dr \tag{16.20}$$

16.2 The Van der Waals equation

Now we turn to a specific example, let $u_0(r)$ takes a particular form

$$u_0(r) = \begin{cases} 0 & r > \sigma \\ \infty & 0 < r \leq \sigma \end{cases} \tag{16.21}$$

This represents a *hard sphere* potential. In the low-density limit, the radial distribution function

$$g(r) = \exp[-\beta u_0(r)] = \begin{cases} 1 & r > \sigma \\ 0 & 0 < r \leq \sigma \end{cases} \tag{16.22}$$

$u_1(r)$ can be some arbitrary attractive potential.

Now, the first term in A^1 is

$$\begin{aligned}A^1 &= \frac{1}{2}\rho^2 V\int_0^\infty 4\pi r^2 u_1(r)g_0(r) \\ &= \frac{1}{2}\rho^2 V\int_\sigma^\infty 4\pi r^2 u_1(r)dr \\ &= -aN\rho\end{aligned} \tag{16.23}$$

where

$$a = -2\pi\int_\sigma^\infty r^2 u_1(r)dr \tag{16.24}$$

is a number dependent on σ and the specific form of $u_1(r)$.

Z_N^0 can be determined analytically. If σ=0,

$$Z_N^0 = V^N \tag{16.25}$$

However, this is not practical. If we consider two hard spheres at closest contact, we can find that the sphere has a radius σ. Hence the excluded volume for these two particles is $\frac{4}{3}\pi\sigma^3$. Let's denote this as b. Then Nb is the total excluded volume, we find that, the partition function is approximately

$$Z_N^0 = (V - Nb)^N \tag{16.26}$$

Thus,

$$A(N, V, T) \approx -\frac{1}{\beta} \ln \left[\frac{(V - Nb)^N}{N!\lambda^{3N}} \right] - \frac{aN^2}{V} \tag{16.27}$$

P can be expressed as

$$\begin{aligned} \frac{P}{kT} &= -\frac{1}{kT} \left(\frac{\partial A}{\partial V} \right)_{N,T} \\ &= \frac{N}{V - Nb} - \frac{aN^2}{kTV^2} \\ &= \frac{\rho}{1 - \rho b} - \frac{a\rho^2}{kT} \end{aligned} \tag{16.28}$$

If we let $v = N/V$,

$$P = \frac{RT}{v - b} - \frac{a}{v^2} \tag{16.29}$$

This is the famous *van de Waals equation*.

16.3 Virial expansion to 2nd order

In the very low density limit, we may assume that $\rho b \ll 1$

$$\frac{1}{1 - \rho b} \approx 1 + \rho b \tag{16.30}$$

$$\frac{P}{kT} \approx \rho + \rho^2 \left(b - \frac{a}{kT} \right) \tag{16.31}$$

From here we can also approximate the 2nd virial coefficient,

$$B_2(T) \approx b - \frac{a}{kT} = \frac{2}{3}\pi\sigma^3 + \frac{2\pi}{kT} \int_\sigma^\infty r^2 u_1(r) dr \tag{16.32}$$